

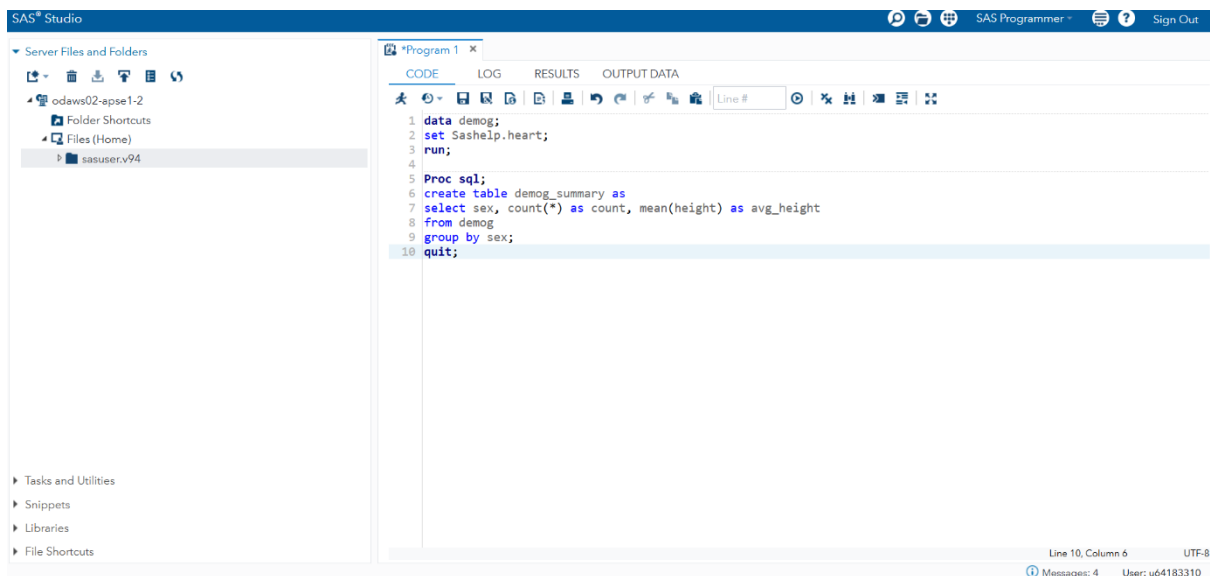
SAS Project Submission

Category-4

Q1) Creating a Demographic Summary Table:

```
1 proc sql;  
2     create table demog_summary as  
3     select gender, count(*) as count, mean(age) as avg_age  
4     from demog  
5     group by gender;  
6 quit;
```

Ans 1:



SAS® Studio

Server Files and Folders

- odaws02-apse1-2
 - Folder Shortcuts
 - Files (Home)
 - sasuser.v94

Tasks and Utilities

- Snippets
- Libraries
- File Shortcuts

*Program 1

CODE LOG RESULTS OUTPUT DATA

Errors, Warnings, Notes

- Errors
- Warnings
- Notes (5)

NOTE: There were 5209 observations read from the data set SASHELP.HEART.
 NOTE: The data set WORK.DEMOG has 5209 observations and 17 variables.
 NOTE: DATA statement used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	2329.81k
OS Memory	22188.00k
Timestamp	14/05/2025 07:34:32 AM
Step Count	43
Page Faults	0
Page Reclaims	249
Page Swaps	0
Voluntary Context Switches	11
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	1800

72
 73 Proc sql;
 74 create table demog_summary as
 75 select sex, count(*) as count, mean(height) as avg_height
 76 from demog
 77 group by sex;

Messages: 4 User: u64183310

14/05/2025, 13:04

Results: WORK.DEMOG_SUMMARY

Obs	Sex	count	avg_height
1	Female	2873	62.5726
2	Male	2336	67.5674

Q2) Merging Datasets:

```

1 proc sql;
2     create table merged_data as
3     select a.*, b.treatment
4     from patients a
5     left join treatment b
6     on a.patient_id = b.patient_id;
7 quit;

```

Ans2:

SAS® Studio

Server Files and Folders
Tasks and Utilities
Snippets
Libraries
My Libraries

File Shortcuts

*Program 1

CODE LOG RESULTS OUTPUT DATA

```
1 data a;  
2 set sasHELP.aacomp(obs=10);  
3 run;  
4  
5 data b;  
6 set sasHELP.aarfm(obs=10);  
7 run;  
8  
9 proc sql;  
10 create table merged_data as  
11 select a.*, b.lineno  
12 from a as a  
13 left join b as b  
14 on a.locale = b.locale;  
15 quit;  
16
```

Line 16, Column 1 UTF-8
Messages: 7 User: u64183310

SAS® Studio

Server Files and Folders
Tasks and Utilities
Snippets
Libraries
My Libraries

File Shortcuts

*Program 1

CODE LOG RESULTS OUTPUT DATA

Errors, Warnings, Notes

Errors (1)
Warnings (1)
Notes (8)

```
82 on a.locale = b.locale;  
WARNING: Variable lineno already exists on file WORK.MERGED_DATA.  
NOTE: Table WORK.MERGED_DATA created, with 100 rows and 4 columns.  
83 quit;  
NOTE: PROCEDURE SQL used (Total process time):  
real time 0.00 seconds  
user cpu time 0.01 seconds  
system cpu time 0.00 seconds  
memory 5900.46k  
OS Memory 27824.00k  
Timestamp 14/05/2025 04:10:20 PM  
Step Count 67 Switch Count 2  
Page Faults 0  
Page Reclaims 261  
Page Swaps 0  
Voluntary Context Switches 12  
Involuntary Context Switches 0  
Block Input Operations 0  
Block Output Operations 288  
84  
85  
86 OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
```

Messages: 7 User: u64183310

Obs	locale	key	lineno	text
1	en	MODEL_CRDEPTH_VLABEL	1	CR Sample Depth
2	en	MODEL_CUMLIFT_VLABEL	1	Cumulative Lift
3	en	MODEL_CODEDEPENDENCY_ERROR	1	Unable to determine the code variable dependencies.
4	en	MODEL_CREATEINTERMEDIATEFILE_ERROR	1	Intermediate file creation failed.
5	en	MODEL_CUMEVENTCOUNT_VLABEL	1	Cumulative Event Count
6	en	MODEL_CRCUT_VLABEL	1	CR Cutoff
7	en	MODEL_ASE_VLABEL	1	Average Squared Error
8	en	MODEL_CCUT_VLABEL	1	Classification Cutoff
9	en	MODEL_CORRECTRATE_VLABEL	1	Overall Precision Rate
10	en	MODEL_CR_VLABEL	1	Maximum Classification Rate (CR)
11	en	MODEL_CRDEPTH_VLABEL	1	CR Sample Depth
12	en	MODEL_CUMLIFT_VLABEL	1	Cumulative Lift
13	en	MODEL_CODEDEPENDENCY_ERROR	1	Unable to determine the code variable dependencies.
14	en	MODEL_CREATEINTERMEDIATEFILE_ERROR	1	Intermediate file creation failed.
15	en	MODEL_CUMEVENTCOUNT_VLABEL	1	Cumulative Event Count
16	en	MODEL_CRCUT_VLABEL	1	CR Cutoff
17	en	MODEL_ASE_VLABEL	1	Average Squared Error
18	en	MODEL_CCUT_VLABEL	1	Classification Cutoff
19	en	MODEL_CORRECTRATE_VLABEL	1	Overall Precision Rate
20	en	MODEL_CR_VLABEL	1	Maximum Classification Rate (CR)
21	en	MODEL_CRDEPTH_VLABEL	1	CR Sample Depth
22	en	MODEL_CUMLIFT_VLABEL	1	Cumulative Lift
23	en	MODEL_CODEDEPENDENCY_ERROR	1	Unable to determine the code variable dependencies.
24	en	MODEL_CREATEINTERMEDIATEFILE_ERROR	1	Intermediate file creation failed.
25	en	MODEL_CUMEVENTCOUNT_VLABEL	1	Cumulative Event Count
26	en	MODEL_CRCUT_VLABEL	1	CR Cutoff
27	en	MODEL_ASE_VLABEL	1	Average Squared Error
28	en	MODEL_CCUT_VLABEL	1	Classification Cutoff
29	en	MODEL_CORRECTRATE_VLABEL	1	Overall Precision Rate
30	en	MODEL_CR_VLABEL	1	Maximum Classification Rate (CR)
31	en	MODEL_CRDEPTH_VLABEL	1	CR Sample Depth
32	en	MODEL_CUMLIFT_VLABEL	1	Cumulative Lift
33	en	MODEL_CODEDEPENDENCY_ERROR	1	Unable to determine the code variable dependencies.

Q3) Filtering Data:

```

1 proc sql;
2     create table filtered_data as
3     select *
4     from lab_results
5     where test_date between '01JAN2023'd and '31DEC2023'd;
6 quit;

```

Ans3:

SAS® Studio

Server Files and Folders

Tasks and Utilities

Libraries

File Shortcuts

*Program 1

CODE

LOG

RESULTS

OUTPUT DATA

Line #

1 data filter;

2 set sashelp.buy;

3 run;

4

5 proc sql;

6 create table filtered_data as

7 select *

8 from filter

9 where date between '01JAN1999'd and '17OCT2005'd;

10 quit;

Line 10, Column 6

UTF-8

Messages: 9

User: u64183310

SAS® Studio

Server Files and Folders

Tasks and Utilities

Libraries

File Shortcuts

*Program 1

CODE

LOG

RESULTS

OUTPUT DATA

Errors, Warnings, Notes

Errors

Warnings

Notes (5)

NOTE: There were 11 observations read from the data set SASHELP.BUY.

NOTE: The data set WORK.FILTER has 11 observations and 2 variables.

NOTE: DATA statement used (Total process time):

real time 0.00 seconds

user cpu time 0.00 seconds

system cpu time 0.01 seconds

memory 829.18k

OS Memory 24744.00k

Timestamp 14/05/2025 04:24:33 PM

Step Count 91 Switch Count 2

Page Faults 0

Page Reclaims 145

Page Swaps 0

Voluntary Context Switches 11

Involuntary Context Switches 0

Block Input Operations 0

Block Output Operations 264

72

73 proc sql;

74 create table filtered_data as

75 select *

76 from filter

77 where date between '01JAN1999'd and '17OCT2005'd;

Messages: 9

User: u64183310

14/05/2025, 21:55

Results: WORK.FILTERED_DATA

Obs	DATE	AMOUNT
1	01JAN1999	-51000
2	01JAN2000	-2000
3	01JAN2001	-2000
4	01JAN2002	-2000
5	01JAN2003	-2000
6	01JAN2004	-2000
7	01JAN2005	-2000

Q4) Calculating Summary Statistics:

```
1 proc sql;
2     create table summary_stats as
3     select treatment, mean(blood_pressure) as avg_bp, std(blood_pressure) as std_bp
4     from vitals
5     group by treatment;
6 quit;
```

Ans4:

SAS® Studio

Server Files and Folders

Tasks and Utilities

Utilities

Snippets

Libraries

File Shortcuts

*Program 1

CODE LOG RESULTS OUTPUT DATA

1 data sum_stat;

2 set sashelp.heart;

3 run;

4

5 proc sql;

6 create table summary_statistics as

7 select status, mean(AgeAtStart) as avg_age_at_strt, std(MRW) as std_MRW

8 from sum_stat

9 group by status;

10 quit;

Line 10, Column 6

UTF-8

SAS® Studio

*Program 1

CODE LOG RESULTS OUTPUT DATA

Errors, Warnings, Notes

Errors

Warnings

Notes (5)

72

73 proc sql;

74 create table summary_statistics as

75 select status, mean(AgeAtStart) as avg_age_at_strt, std(MRW) as std_MRW

76 from sum_stat

77 group by status;

NOTE: Table WORK.SUMMARY_STATISTICS created, with 2 rows and 3 columns.

78 quit;

NOTE: PROCEDURE SQL used (Total process time):

real time 0.00 seconds

user cpu time 0.01 seconds

system cpu time 0.01 seconds

memory 6401.40k

OS Memory 30764.00k

Timestamp 14/05/2025 04:34:49 PM

Step Count 118 Switch Count 2

Page Faults 0

Page Reclaims 379

Page Swaps 0

Voluntary Context Switches 11

Involuntary Context Switches 2

Block Input Operations 0

Block Output Operations 280

14/05/2025, 22:05

Results: WORK.SUMMARY_STATISTICS

Obs	Status	avg_age_at_strt	std_MRW
1	Alive	41.1311	18.6859
2	Dead	48.8167	21.5416

Q5) Creating Patient Visit Count:

```
1 proc sql;
2     create table visit_count as
3     select patient_id, count(*) as visit_count
4     from visits
5     group by patient_id;
6 quit;
7
```

Ans5:

SAS Studio interface showing the CODE tab. The code is as follows:

```
1 data visits;
2 set Sashelp.bweight;
3 run;
4
5 proc sql;
6 create table visit_count as
7 select boy, count(*) as boys_or_girls_visit_count
8 from visits
9 group by boy;
10 quit;
```

Line 2, Column 21 UTF-8
Messages: 31 User: u64183310

SAS Studio interface showing the LOG tab. The log contains the following information:

```
NOTE: There were 50000 observations read from the data set SASHELP.BWEIGHT.
NOTE: The data set WORK.VISITS has 50000 observations and 10 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.01 seconds
      system cpu time     0.01 seconds
      memory              2784.31k
      OS Memory           27564.00k
      Timestamp           14/05/2025 04:51:54 PM
      Step Count          261  Switch Count   3
      Page Faults         0
      Page Reclaims       335
      Page Swaps          0
      Voluntary Context Switches 21
      Involuntary Context Switches 0
      Block Input Operations 976
      Block Output Operations 7944
```

72
73
74
75
76
77

```
proc sql;
create table visit_count as
select boy, count(*) as boys_or_girls_visit_count
from visits
group by boy;
```

Messages: 31 User: u64183310

14/05/2025, 22:22

Results: WORK.VISIT_COUNT

Obs	Boy	boys_or_girls_visit_count
1	0	24208
2	1	25792

In above example '0' means Girl and '1' means Boy. So 25,792 boy patients and 24,208 girl patients visited the hospital.

Q6) Identify Missing Data:

```
1 proc sql;
2     create table missing_data as
3     select *
4     from lab_results
5     where result is missing;
6 quit;
7
```

Ans6:

The screenshot displays the SAS Studio interface. On the left, the 'Server Files and Folders' pane shows a project structure with folders like 'odaws01-apse1-2' and 'sasuser.v94', and a file 'sasmovies.xlsx'. The main workspace is divided into three tabs: 'CODE', 'LOG', and 'RESULTS'. The 'CODE' tab is active, showing a SAS program that creates a table named 'missing_data' from the 'lab_results' table, selecting all rows where the 'result' is missing. The program is as follows:

```
1 proc sql;
2     create table missing_data as
3     select *
4     from lab_results
5     where result is missing;
6 quit;
```

The status bar at the bottom indicates 'Line 4, Column 18' and 'UTF-8'. The bottom right corner shows 'Messages: 42' and 'User: u04183310'.

SAS® Studio

Server Files and Folders

- odaws01-apse1-2
 - Folder Shortcuts
 - Files (Home)
 - sasuser.v94
 - sasmovies.xlsx

Tasks and Utilities

- Snippets
- Libraries
- File Shortcuts

*Program 1 x *sasmovies x

CODE LOG RESULTS OUTPUT DATA

Errors, Warnings, Notes

Errors

Warnings

Notes (2)

```

1  OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
68
69  proc sql;
70    create table missing_data as
71    select *
72    from work.import1
73    where Title is missing;
NOTE: Table WORK.MISSING_DATA created, with 9 rows and 8 columns.
74  quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              5775.43k
      OS Memory           31660.00k
      Timestamp           14/05/2025 05:18:44 PM
      Step Count          348  Switch Count  2
      Page Faults         0
      Page Reclaims       191
      Page Swaps          0
      Voluntary Context Switches 12
      Involuntary Context Switches 0
      Block Input Operations 0
      Block Output Operations 264
  
```

Messages: 42 User: u64183310

14/05/2025, 22:51

Results: WORK.MISSING_DATA

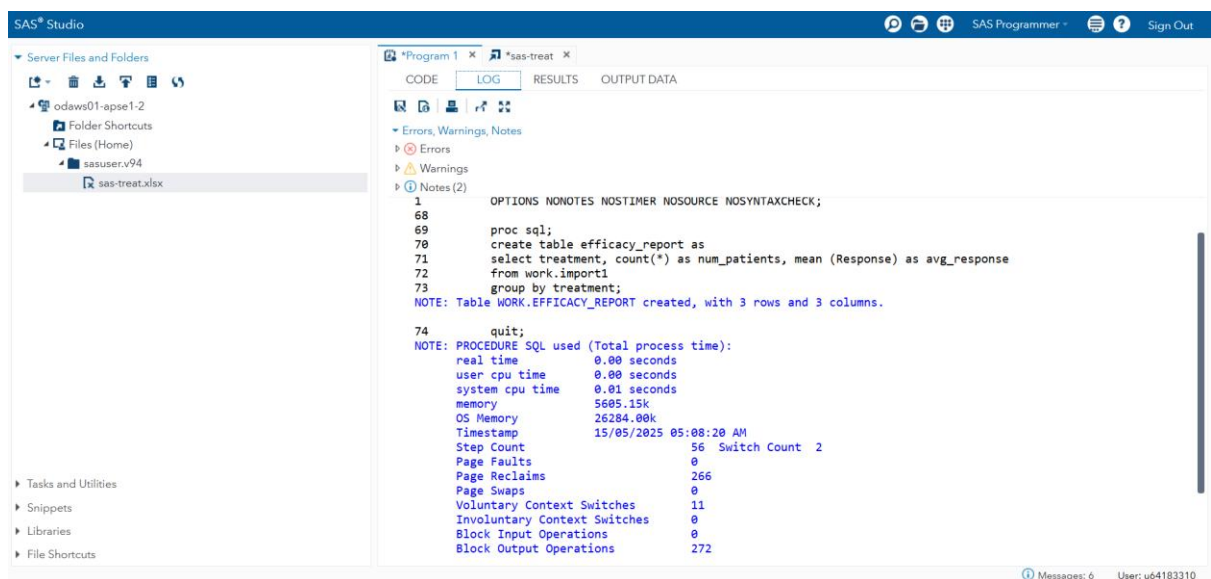
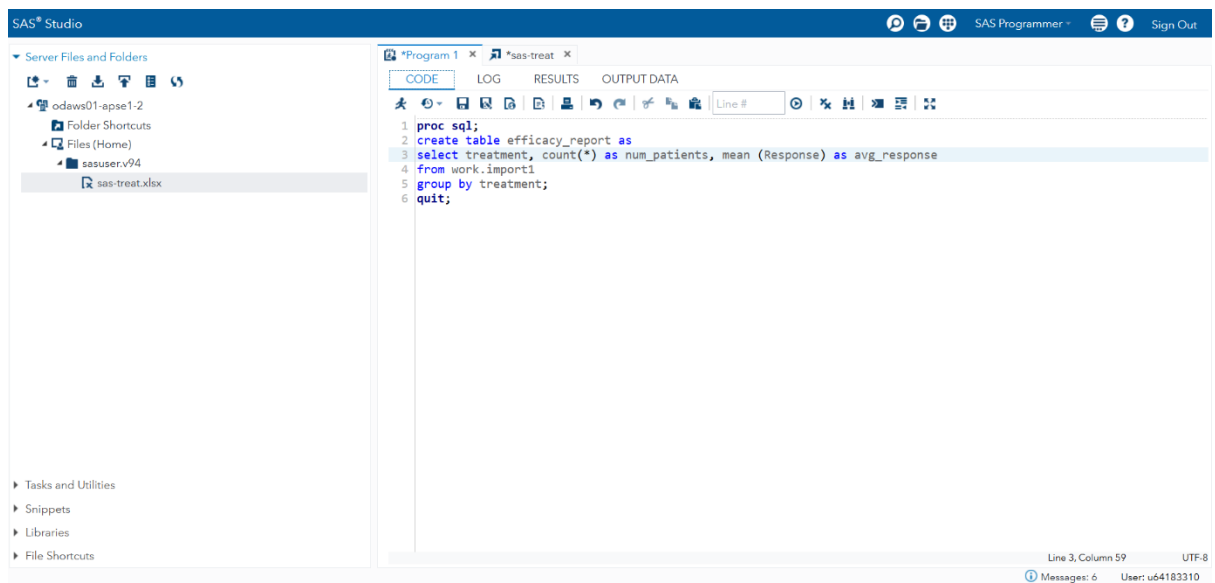
Obs	A	Title	Length	Category	Year	Studio	Rating
1	2		103	Drama	1942	MGM / UA	PG
2	4		116	Comedy	1988	Paramount Pictures	PG-13
3	7		142	Drama	1994	Paramount Pictures	PG-13
4	10		127	Action	1993	Universal Pictures	PG-13
5	12		.	Drama	1996	Warner Brothers	PG
6	14		115	Drama	1982	Paramount Pictures	PG
7	18		124	Sci-Fi	1977	Lucas Film Ltd	R
8	21		101	Adventure	1939	MGM / UA	G
9	26		102	Adventure	1939	MGM / UA	G

Q7) Generating a Treatment Efficacy Report:

```

1  proc sql;
2    create table efficacy_report as
3    select treatment, count(*) as num_patients, mean(response) as avg_response
4    from efficacy
5    group by treatment;
6  quit;
7
  
```

Ans7:



15/05/2025, 10:39

Results: WORK.EFFICACY_REPORT

Obs	Treatment	num_patients	avg_response
1	Heart Treatment	10	84.4
2	Kidney Treatment	10	69.0
3	Lung Treatment	10	53.4

Q8) Creating an Adverse Events Summary:

```
1 proc sql;
2     create table ae_summary as
3     select event_type, count(*) as num_events
4     from adverse_events
5     group by event_type;
6 quit;
7
```

Ans8:

The screenshot displays the SAS Studio interface. The left pane shows the 'Server Files and Folders' tree with the file 'sas-adverse.xlsx' selected. The main editor pane shows the following SAS code:

```
1 proc sql;
2     create table adverse_events_summary as
3     select adverse_events, count(*) as num_events
4     from work.import
5     group by adverse_events;
6 quit;
```

The 'LOG' tab is active, showing the execution output. The output includes the following information:

```
1      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
68
69      proc sql;
70      create table adverse_events_summary as
71      select adverse_events, count(*) as num_events
72      from work.import
73      group by adverse_events;
NOTE: Table WORK.ADVARSE_EVENTS_SUMMARY created, with 26 rows and 2 columns.
74      quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time          0.00 seconds
      user cpu time      0.00 seconds
      system cpu time    0.00 seconds
      memory             5603.53k
      OS Memory          26540.00k
      Timestamp          15/05/2025 05:22:21 AM
      Step Count         75   Switch Count  2
      Page Faults        0
      Page Reclaims     245
      Page Swaps         0
      Voluntary Context Switches 14
      Involuntary Context Switches 1
      Block Input Operations 0
      Block Output Operations 272
```

Obs	Adverse_Events	num_events
1	Abdominal Pain	2
2	Anemia	1
3	Anxiety	1
4	Back Pain	2
5	Chest Pain	2
6	Constipation	2
7	Cough	2
8	Depression	1
9	Diarrhea	2
10	Dizziness	3
11	Fatigue	3
12	Fever	2
13	Headache	3
14	Hypertension	2
15	Hypotension	2
16	Insomnia	2
17	Itching	1
18	Joint Pain	1
19	Muscle Pain	1
20	Nausea	3
21	Palpitations	2
22	Rash	3
23	Shortness of Breath	2
24	Sweating	1
25	Tremor	2
26	Vomiting	3

Q9) Listing Patients with Specific Conditions:

```
1 proc sql;  
2   create table specific_conditions as  
3   select patient_id, condition  
4   from conditions  
5   where condition in ('Diabetes', 'Hypertension');  
6 quit;  
7
```

Ans9:

The screenshot shows the SAS Studio interface. On the left, the 'Server Files and Folders' pane displays a tree structure with 'odaws01-apse1-2' as the root, containing 'Folder Shortcuts', 'Files (Home)', 'sasuser.v94', and 'sas-cond.xlsx'. The main editor area has tabs for '*sas-cond' and '*Program 1'. The 'CODE' tab is active, showing the following SAS code:

```
1 proc sql;  
2 create table specific_conditions as  
3 select patient_id, condition  
4 from work.import1  
5 where condition in ('Diabetes', 'Hypertension');  
6 quit;
```

At the bottom right, the status bar indicates 'Line 4, Column 18' and 'UTF-8'. Below the status bar, it shows 'Messages: 12' and 'User: u64183310'.

The screenshot shows the SAS Studio interface with the 'LOG' tab active. The left pane is the same as in the previous screenshot. The main editor area displays the execution log for the PROC SQL procedure:

```
1      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;  
68  
69      proc sql;  
70      create table specific_conditions as  
71      select patient_id, condition  
72      from work.import1  
73      where condition in ('Diabetes', 'Hypertension');  
NOTE: Table WORK.SPECIFIC_CONDITIONS created, with 24 rows and 2 columns.  
74      quit;  
NOTE: PROCEDURE SQL used (Total process time):  
      real time           0.00 seconds  
      user cpu time       0.01 seconds  
      system cpu time     0.00 seconds  
      memory              5695.56k  
      OS Memory          26796.00k  
      Timestamp           15/05/2025 05:37:47 AM  
      Step Count          188  Switch Count  2  
      Page Faults         0  
      Page Reclaims       195  
      Page Swaps          0  
      Voluntary Context Switches 11  
      Involuntary Context Switches 1  
      Block Input Operations 0  
      Block Output Operations 264
```

At the bottom right, the status bar indicates 'Messages: 12' and 'User: u64183310'.

Obs	patient_id	condition
1	P001	Diabetes
2	P002	Hypertension
3	P006	Diabetes
4	P007	Hypertension
5	P011	Diabetes
6	P012	Hypertension
7	P016	Diabetes
8	P017	Hypertension
9	P021	Diabetes
10	P022	Hypertension
11	P026	Diabetes
12	P027	Hypertension
13	P030	Diabetes
14	P031	Hypertension
15	P035	Diabetes
16	P036	Hypertension
17	P043	Diabetes
18	P044	Hypertension
19	P050	Diabetes
20	P051	Hypertension
21	P061	Diabetes
22	P062	Hypertension
23	P072	Diabetes
24	P073	Hypertension

Q10) Calculating Time to Event:

```
1 proc sql;  
2     create table time_to_event as  
3     select patient_id, event_date - start_date as time_to_event  
4     from events;  
5 quit;  
6
```

Ans10)

The screenshot shows the SAS Studio interface. The left pane displays the 'Server Files and Folders' tree with a file named 'sas_event_days.xlsx' selected. The main editor area is titled '*sas_event_days' and '*Program 1'. The 'CODE' tab is active, showing the following SAS code:

```
1 proc sql;  
2 create table time_to_event as  
3 select patient_id, event_date - start_date as time_to_event  
4 from work.import;  
5 quit;
```

The bottom status bar indicates 'Line 5, Column 6' and 'UTF-8'. The message bar shows 'Messages: 2' and 'User: u04183310'.

The screenshot shows the same SAS Studio interface, but the 'LOG' tab is active. It displays the execution results of the PROC SQL query. The log includes the following text:

```
1      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;  
68  
69      proc sql;  
70      create table time_to_event as  
71      select patient_id, event_date - start_date as time_to_event  
72      from work.import;  
NOTE: Table WORK.TIME_TO_EVENT created, with 30 rows and 2 columns.  
73      quit;  
NOTE: PROCEDURE SQL used (Total process time):  
      real time          0.00 seconds  
      user cpu time      0.00 seconds  
      system cpu time    0.00 seconds  
      memory             5602.71k  
      OS Memory          26284.00k  
      Timestamp          15/05/2025 07:57:05 AM  
      Step Count         31      Switch Count  2  
      Page Faults        0  
      Page Reclaims      210  
      Page Swaps          0  
      Voluntary Context Switches 11  
      Involuntary Context Switches 0  
      Block Input Operations 0  
      Block Output Operations 264
```

The bottom status bar shows 'Messages: 2' and 'User: u04183310'.

15/05/2025, 13:27

Results: WORK.TIME_TO_EVENT

Obs	patient_id	time_to_event
1	P1000	10
2	P1001	1
3	P1002	11
4	P1003	7
5	P1004	4
6	P1005	6
7	P1006	0
8	P1007	8
9	P1008	3
10	P1009	9
11	P1010	2
12	P1011	3
13	P1012	6
14	P1013	11
15	P1014	6
16	P1015	15
17	P1016	5
18	P1017	0
19	P1018	4
20	P1019	8
21	P1020	14
22	P1021	2
23	P1022	12
24	P1023	8
25	P1024	9
26	P1025	2
27	P1026	10
28	P1027	12
29	P1028	2
30	P1029	15

---THE END---